

Interface Capability Brief

EDI 270/271 Eligibility · Stakeholder overview · 2026-08-13

1. Executive summary

Clinic front desk verifies member eligibility over real X12 270/271 (005010X279A1): build a 270, call a mock payer, parse the 271, and show AAA rejection theater then a successful EB benefits response. Demo only — not a real payer.

2. Who this is for

- Sources: Clinic Web UI posts EligibilityRequest XML - Destinations: sync X12 270, mock-payer 271, clinic JSON summary; audit files under output/ - Demo vs production: Demo only — canned AAA/EB, no SNIP, no clearinghouse

- Sources: Clinic Web UI posts EligibilityRequest XML
- Destinations: sync X12 270, mock-payer 271, clinic JSON summary; audit files under output/
- Demo vs production: Demo only — canned AAA/EB, no SNIP, no clearinghouse

3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: RESTful Web Service (SERVICE_NAME=eligibility, resources build / parse, Synchronous=true)
- Format / envelope: XML request → PilotFish EDI XML → X12 270; raw 271 wrapped as <EdiPayload> for parse
- Identity fields: MemberId (NM1IL...MI), TraceNumber (TRN02 / BHT03)
- Samples path: TableData examples in samples/; live story uses UI presets FAIL001 / OK001 / UNKNOWN

Integration highlights

FQCN sources: CRL Plus REST listener, edi-835-payment-integrity EDI transform + XML Formatting, PilotFish Documentation 26R1.11. Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
Listener	com.pilotfish.eip.modules.http.rest.RESTfulWebServiceListener	Skeleton: CRL Plus Status.91 REST listener (POST, no auth). Synchronous=true because this is request/response, not a queue.
Router	XPathRoutingModule OGNL on HttpMethodName + ResourceName	build / parse / 405
Build processors	XSLT → XML Formatting → EDITransformationProcessor XML→EDI	TableData edi-tabledata/270-A1 (5010). XML Formatting on the 270 EDI XML audit write, not the listener.
Parse processors	File snapshot → XSLT structural 271→JSON	23R1 EDI→XML for 271 has failed without licensed tables; parse uses wrapped raw 271. TableData 271-A1 is still mounted.
Transports	SynchronousResponseTransport	Returns 270 text or JSON
Mock payer	Flask :8210/x12/270	FAIL001→AAA 72, UNKNOWN→AAA 75, else success EB

4. Honest boundaries (not claimed yet)

-
- Switch parse to EDITransformationProcessor EDI→XML once 271 TableData is proven on 23R1?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Eligibility 270 271 API” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: 270 has ISA/ST; 271 has AAA or EB; clinic banner from parse JSON - What is NOT checked: SNIP 270/271, real payer connectivity, HIPAA code-set licensing - Does failure block outbound? yes for unparseable request XML

6. State & idempotency

- Status model: none (sync request/response) - Dedup keys: TraceNumber on the wire - Retry / poison: bad XML fails at XSLT; unknown methods 405

8. Dual-write / side effects

- Order: sync response is the contract; file snapshots are audit only - Compensation: none - Demo shortcuts: UI may insert ST*270 if XML→EDI omitted it; UI wraps 271 in EdiPayload for parse

3. Inbound contract

- Transport: RESTful Web Service (SERVICE_NAME=eligibility, resources build / parse, Synchronous=true) - Format / envelope: XML request → PilotFish EDI XML → X12 270; raw 271 wrapped as <EdiPayload> for parse - Identity fields: MemberId (NM1IL...MI), TraceNumber (TRN02 / BHT03) - Samples path: TableData examples in samples/; live story uses UI presets FAIL001 / OK001 / UNKNOWN

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Conditional Router, Return 270, Return Summary.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Return 270.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

1 - Eligibility 270 271 API: runs steps such as Eligibility REST, Snapshot Inbound Body, Conditional Router, Map Request To 270 EDI XML.

7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — Mock payer is not a real 271 engine — Pitch is eligibility theater
- Risk noted in design: Med — 23R1 trial X12 tables expired — Sandbox mounts EDI/TableData/x12 → eip-root/edi-tabledata
- Risk noted in design: Med — Duplicate processor names — EIP will not load the route
- Risk noted in design: Low — Clinic UI orchestrates three HTTP hops — Sync REST + HttpPost mismatch

8. Seeing it work

- - Ports: Mock payer 8210, EIP 8106, Web UI 8107 - Volumes: ./output, ./logs, ./samples, ./documents, TableData mount (5 ../ from Insurance/EDI/) - Heap: 512M–2GB - Dependencies / cold start: EIP ~60–90s - Compose project: edi-270-271-eligibility
- Local: http://127.0.0.1:8107/
- EIP: http://127.0.0.1:8106/eip/
- Eligibility API: http://127.0.0.1:8106/eip/rest/eligibility/
- Mock payer: http://127.0.0.1:8210/x12/270
- docker compose --profile full up -d --build

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.